General User Manual for Electronic Components

1. Product Overview

This series of electronic components is a high-precision, high-stability electronic accessory widely applied in electronic DIY projects, industrial control equipment, smart home systems, experimental testing devices and other electronic products. The product features compact structure, low power consumption, stable performance and strong compatibility, which can perfectly meet the daily use and commercial matching needs of most electronic equipment.

2. General Specifications

- **Working Voltage**: 3.0V - 5.5V DC (universal for most single-chip systems)
- **Working Current**: Low power consumption, standby micro-current
- **Working Temperature**: -20°C ~ +85°C (conventional industrial temperature range)
- **Storage Temperature**: -40°C ~ +125°C
- **Compatibility**: Compatible with Arduino, Raspberry Pi, STM32 and other mainstream single-chip microcomputer systems
- **Performance Features**: High stability, anti-interference, high precision, long service life
- **Insulation Property**: Environmentally friendly insulating material, safe and reliable

3. Installation & Operation Instructions

1. Confirm the product model and interface definition before installation to avoid wrong connection.
2. Cut off the power supply of the main control board before wiring installation to prevent component burnout caused by live operation.
3. Connect the pins strictly according to the product wiring definition (power pin, signal pin, ground pin).
4. Fix the product on a flat and dry installation position, avoid long-term exposure to humid, high temperature and corrosive environments.
5. After wiring, check the circuit connection again, and power on for test after confirming no error.

6. Complete corresponding program configuration according to the product usage requirements to realize normal functional output.

4. Precautions

- **Voltage Protection**: Do not exceed the rated working voltage to avoid permanent damage to the component.
- **Anti-static**: Static electricity may damage precision electronic components. Please take anti-static measures during installation and operation.
- **Waterproof & Moistureproof**: Conventional models are non-waterproof. Please avoid water ingress and humid environment use.
- **Welding Specification**: The welding temperature shall not exceed 350°C, and the single welding time shall be controlled within 3 seconds to avoid damaging the chip and pins.
- **Reverse Connection Prevention**: Strictly prohibit reverse connection of positive and negative power supply, otherwise the product will be burned out.
- **Overload Prohibition**: Do not overload use beyond the product parameter range.

5. Maintenance & Daily Care

- Keep the product surface clean and dry, regularly remove dust and foreign matters on the surface.
- Check the wiring firmness regularly to avoid poor contact caused by loose wiring.
- Long-term unused equipment shall be stored in a dry, ventilated and dust-proof environment.
- If the product fails, stop using it immediately and do not disassemble or repair it by yourself without authorization.

6. Environmental & Safety Standard

This product complies with EU RoHS 2.0 (2011/65/EU & EU 2015/863) environmental protection standards, and does not contain harmful and prohibited substances. It meets the requirements of low voltage safety and electromagnetic compatibility directives, and is safe for commercial sales and global use.

7. After-sales Service

- We provide long-term after-sales support and technical consultation.
- If there is any quality problem with the product during normal use, we support replacement and after-sales processing.

- Please contact our customer service in time for any product usage problems.

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（注：部分内容可能由 AI 生成）